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Original article

Economic impact of severe and non-severe hypoglycemia in patients with Type 1 and Type 2 diabetes in the United States

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Citation: J Med Econ 2015; 18:420–32**Abstract****Objective:**

To identify the direct and indirect costs of hypoglycemia in patients with Type 1 or Type 2 diabetes mellitus (DM) in the US setting.

Methods:

A literature review was conducted to identify and review studies that reported data on the economic burden of hypoglycemia and the related medical resource consumption or productivity loss related to hypoglycemia in patients with Type 1 or Type 2 DM. Relevant information was collated in an economic model to assess the direct and indirect costs following severe and non-severe hypoglycemic events in Type 1 and Type 2 DM.

Results:

Detailed evidence of the medical cost burden of hypoglycemic events was identified from 14 studies. For both Type 1 and Type 2 DM, episodes requiring assistance from a healthcare practitioner were identified as particularly costly and amounted to \$1161 per episode (direct costs) compared with episode costs of \$66 and \$11 for events requiring third-party (non-medical) assistance and events managed by self-treatment, respectively. Indirect costs associated with severe hypoglycemia requiring non-medical assistance, severe hypoglycemia requiring medical assistance, and non-severe hypoglycemia were predicted to be \$242, \$160, and \$11 for patients with Type 1 diabetes and \$579, \$176, and \$11 for patients with Type 2 diabetes, respectively.

Conclusion:

Both severe and non-severe hypoglycemia incur substantial healthcare costs. Failure to account for these costs may under-estimate the value of management strategies that minimize hypoglycemia risk.

Introduction

Effective blood glucose control is the foundation of diabetes management aimed at minimizing microvascular and macrovascular complications¹. The risk of hypoglycemia increases when glucose control is intensified and this can act as a barrier to achieving blood glucose targets². Hypoglycemia occurs more frequently in type 1 diabetes mellitus (T1DM), but is also a concern in type 2 diabetes mellitus (T2DM) from both a clinical and patient perspective³. The clinical consequences of hypoglycemia have been widely debated following results from recent clinical trials such as ACCORD⁴, which found that the frequency of both non-severe and severe hypoglycemia was considerably higher in the more intensively controlled cohort. Non-severe hypoglycemia events (NSHEs) are known to occur with a greater frequency than severe

hypoglycemia events (SHEs); the UK Hypoglycemia Study Group⁵ reported incidence rates for SHEs of 110 and 320 events per 100 patient-years for individuals with T1DM who were receiving insulin treatment for less than 5 years or more than 15 years, respectively. This compared with 10 and 70 SHEs per 100 patient-years for individuals with T2DM who were receiving insulin treatment for less than 2 years or more than 5 years. In addition, NSHEs were found to be more frequent (410 and 1020 events per 100 patient-years, respectively) among individuals with T2DM within the same respective groups⁵.

Although NSHEs are more frequent, research has focused on the evaluation of the economic and clinical impact of SHEs; the health economic implications of NSHEs are less well understood, with the exception of the Brod *et al.*⁶ paper. A recent survey of 1653 healthcare practitioners (HCPs) and 3587 patients with T2DM treated with insulin reported that NSHEs materially impact resource utilization⁷. A total of 52% of HCPs reported being contacted at least once per month by patients who experienced self-treated hypoglycemia⁷. Furthermore, Brod *et al.*⁶ conducted a multi-country survey including 1404 respondents with T1DM and T2DM to investigate the impact of NSHEs on work productivity and diabetes management. The study reported that 13.7% of US patients contacted a HCP subsequent to a NSHE. Lost productivity associated with NSHEs was estimated to range from \$15.26–\$93.47 (USD) per event, representing 8.3–15.9 hours of lost work time per month⁶.

The cost of hypoglycemia is a complex issue because the burden may occur in multiple ways. Direct medical costs are incurred when healthcare resources are consumed to treat hypoglycemia; indirect medical costs accrue due to lost productivity when patients are absent from work. While there is a body of evidence in the literature regarding the costs and impact associated with hypoglycemia, the information is fragmented and does not include all types of resource utilization associated with both SHEs and NSHEs. Accordingly, the purpose of this study was to identify available data sources and collate such information in an economic model to estimate the direct and indirect cost consequences of both SHEs and NSHEs more precisely from a US payer perspective. These event-based costs can subsequently be applied in a hypoglycemia 'decision tree' model or a diabetes simulation model that considers the respective severity grades of hypoglycemia (e.g., the IMS CORE Diabetes Model^{8,9}).

Methods

Literature review

A detailed pragmatic literature review was performed to identify studies published between June 1, 2007 and June

1, 2012 that investigated or reported the types of medical resources used to treat hypoglycemia events with different severity grades and further the proportion of events (frequency) requiring treatment by these resources. In addition, information on productivity loss associated with SHEs and NSHEs was searched. A review of the MEDLINE database was performed using PubMed. The full search strategy can be found in the online supplemental appendix. Articles were included in the review if they reported the types of medical resources used to treat hypoglycemia events with different severity grades and further the proportion of events (frequency) requiring treatment by these resources. In addition, studies reporting productivity loss associated with SHEs and NSHEs were included. We excluded case reports, reader comments, or author responses, studies performed exclusively in critical care settings, studies that focused on drug poisoning or overdose-related hypoglycemia, non-diabetes conditions that may result in hypoglycemia (e.g., malaria, tumors), and neonatal or gestational hypoglycemia.

The initial number of abstracts identified from the literature search was 893. Following initial abstract review, 862 articles were excluded. Thirty-one articles of potential interest that meet inclusion criteria were included for full text review. In addition, 20 articles that were not identified via the electronic search but were retrieved from the reference sections of identified articles were included.

Information related to healthcare resource consumption potentially involved in treating hypoglycemia of different severity grades and productivity loss associated with hypoglycemia was collected and reviewed. Although no formal quantitative analyses were undertaken, we selected studies based on the quality of design, population characteristics, and size of study, definition of hypoglycemia (which varied from study to study), country setting (US preference), and date of publication. Medical resources that were found to be relevant from an economic perspective for the treatment of hypoglycemia were identified and information on the related frequency with which these resource categories are required to treat hypoglycemia events with different severity grades (NSHEs and SHEs) were collected.

Following the qualitative review, 14 articles were selected to inform the medical cost burden of hypoglycemic events (11 articles informing relevant healthcare resources^{10–20}, three articles informing frequency of resource use^{6,12,13}, two articles informing productivity loss^{6,21}, and two articles informing the frequency of day-time and nocturnal hypoglycemic episodes^{6,22}).

Definition of hypoglycemia

The definition of hypoglycemia varied between articles; we based our definition on what was found to be most

commonly used in diabetes research. Hypoglycemia was sub-classified into NSHEs and SHEs. A NSHE was defined as an event not requiring assistance from another person (self-manageable) and a SHE was defined as an event requiring assistance of another person to actively administer carbohydrates or glucagon or to take other corrective actions. These definitions align with a recent consensus statement of the American Diabetes Association and the European Association for the Study of Diabetes²³ and a consensus achieved between the American Diabetes Association and the Endocrine Society²⁴. For costing purposes, SHEs were further sub-classified into events requiring non-medical, third-party assistance (SHE1) and events requiring medical assistance (SHE2).

Cost calculations

An economic model was developed in Microsoft Excel to estimate the expected direct and indirect event costs associated with NSHEs, SHE1, and SHE2 in 2012 US dollars. Wherever possible, direct and indirect costs were separately evaluated for T1DM and T2DM. However, in most cases, the available data used for cost calculations were obtained from studies that included individuals with both T1DM and T2DM, but no data stratification by diabetes type. Hence, with the exception of indirect costs for SHE1 and SHE2, which were evaluated separately for individuals with T1DM and T2DM, all direct hypoglycemia episode costs and indirect costs for NSHEs are generic and applicable to both diabetes types. All costs were standardized to 2012 using the Medical Care Consumer Price Index²⁵.

Direct cost calculation

To estimate individual cost weights for the direct event cost calculation, the unit costs of all relevant medical resources identified to treat hypoglycemia were multiplied by the respective frequency with which that medical service was required to treat the event. For example, a frequency of 24% (chance that a SHE2 is associated with hospitalization¹²) is multiplied with presumed hospitalization costs of \$5000 to yield a cost weight of \$1200 for hospitalization following SHE2. The sum of individual cost weights (unit cost multiplied by frequency) across all included resource items was calculated to assess the overall direct hypoglycemic event costs separately for NSHEs, SHE1, and SHE2. The direct costs calculation model is illustrated schematically in Figure 1.

Medical resources

Medical resources that were found to be relevant from an economic perspective were extracted from 11 studies^{10–20} and included inpatient stays with and without prior emergency room (ER) visit, ER visits without subsequent hospitalization, ambulance service and outpatient visits in secondary care and visit to a general practitioner (GP) office, GP visit at home, nurse treatment, and telephone contact with medical care in primary care. Additional costs for treatment from non-HCPs such as glucose injections by non-medical personnel or self-treatment costs were considered, as well as costs for self-monitoring blood glucose (SMBG). These resources were considered to reflect the direct medical costs of NSHEs and SHEs.

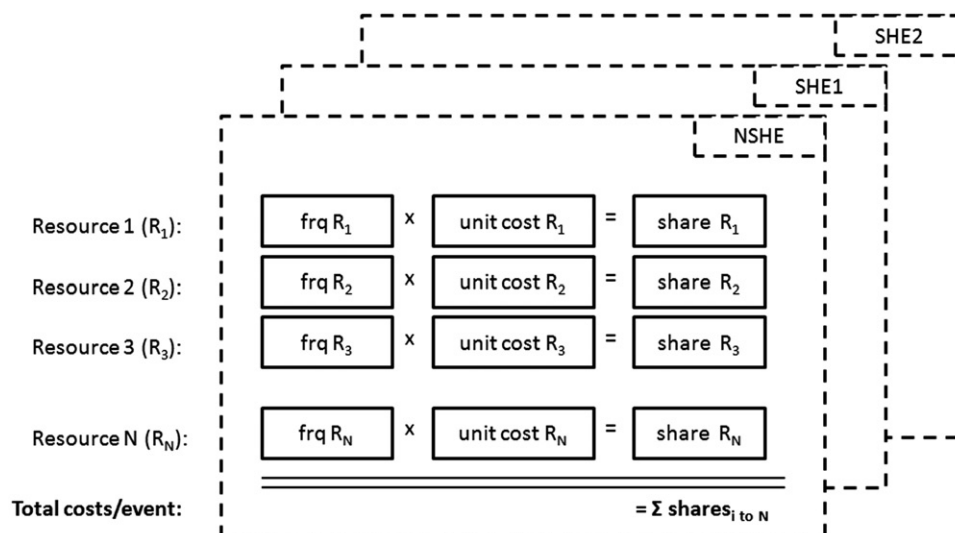


Figure 1. Direct cost calculation model. R_1 to R_N = resource 1 to resource N; $\text{freq } R_1$ to $\text{freq } R_N$ = frequency of resource utilization 1 to N.

Table 1. Unit costs for medical resources included in the hypoglycemia model.

Resource type	Costs	Comment	Source
Secondary care			
ER visit	\$43.67	For physician – 99283 – emergency department visit – medium intensity	Medicare 2012 Physician Fee Schedule
	\$139.14	For the hospital – Ambulatory Payment Classification (APC) 614	Addendum B Medicare Outpatient Prospective Payment System for hospital outpatient services
	\$182.81	Total fee from payer perspective for ER visit (sum of two costs above)	
Ambulance	\$214.47	Base rate for an ambulance (Basic Life Support) averaged across state localities	Medicare ambulance fee schedule
Inpatient	\$7354.00	Diabetes with MCC	Medicare inpatient Prospective Payment System, FY2012 FR, Table 5 Weights
	\$4171.00	Diabetes with CC	
	\$2811.00	Diabetes without CC/MCC	
Inpatient with ER	See inpatient	Inpatient with ER is equal to inpatient (from a payer perspective). The ER visit would be folded into the subsequent inpatient admission.	
Outpatient	\$324.81	US insurance claims data	Curkendall <i>et al.</i> ¹⁴ ; \$285 inflated to 2012
Primary care			
Visit to GP	\$76.23	99203 – physician visit for a medium, new patient	Medicare 2012 Physician Fee Schedule
	\$51.07	99213 – physician visit for medium, established patient	
Home visit from GP	\$93.26	99343 – physician visit at home for medium, new patient	
	\$59.46	99348 – physician visit at home for a medium, established patient	
Nurse	\$43.41	It is assumed that Nurse Practitioners or Clinical Nurse Specialists are providing these services. They are reimbursed at 85% of the physician fee	Medicare 2012 Physician Fee Schedule (85% of 99213 – physician visit for medium, established patient)
Telephone contact with medical care	\$9.87	99441 Phone e/m by phys 5–10 min – \$9.87	Medicare 2012 Physician Fee Schedule
	\$19.00	99442 Phone e/m by phys 11–20 min – \$19.00	
	\$28.13	99443 Phone e/m by phys 21–30 min – \$28.13	
Other			
SMBG	\$1.05	Average cost per testing strip	Yeaw <i>et al.</i> ³⁴ ; \$0.98 inflated to 2012
Non HCP treatment (third-party assistance)	\$55.02	Costs for intramuscular glucagon injection	US net effective sales price of replacement vials
Non HCP treatment (self-treatment)	\$9.23	Extra/special groceries, food and drink and glucose products required following NSHE	Brod <i>et al.</i> ⁶

ER, emergency room; GP, general practitioner; HCP, healthcare practitioner; SMBG, self-monitoring of blood glucose; MCC, major complications and comorbidities; CC, complications and comorbidities; e/m, evaluation and management; NSHE, non-severe hypoglycemic event.

Unit costs

The unit costs of healthcare resources were obtained from US public sources, including Medicare Physician Fee Schedule, Medicare Outpatient Prospective Payment System, Medicare Inpatient Prospective Payment System, and Medicare ambulance fee schedule. Costs for medical resources were sub-classified into primary and secondary care. A summary of the unit cost collection is presented in Table 1.

Primary care unit costs

Primary care costs are based on the US Medicare 2012 Physician Fee Schedule. Primary care physician visit

costs were based on the average cost for an established patient (\$51.07) (i.e., we assumed no home visits by PCPs in the US, but the hypoglycemia model was created to take this possibility into account for future model adaptation). It was assumed that services provided by nurse practitioners or clinical nurse specialists would be reimbursed at 85% of the physician visit fee for a medium, established patient.

Secondary care unit costs

Inpatient costs were obtained from the Medicare Inpatient Prospective Payment System. Although reimbursement rates are separately provided for different complication

levels, the resource cost applied for hypoglycemia-related hospitalization was the cost of inpatient admission associated with complications and comorbidities. Costs for inpatient admission with prior ER visit are assumed to be equal to inpatient costs (from a payer perspective) because the ER visit cost would be folded into the subsequent inpatient admission. Costs for ER visit with no subsequent inpatient admission are calculated on the basis of the costs for the physician (from Medicare 2012 Physician Fee Schedule) and for the hospital ambulatory payment, which were obtained from the Addendum B of Medicare Outpatient Prospective Payment System for hospital outpatient services. Costs for an outpatient visit were not obtained as a standard reimbursable cost because this would require a detailed description of the procedures undertaken in outpatient treatment. Rather, outpatient costs were obtained from Curkendall *et al.*¹⁴, who reported average costs for hypoglycemia-related outpatient treatment according to the analyzed insurance claims data. Ambulance-related costs were obtained from the 2012 Medicare ambulance fee schedule, where the base rate for an ambulance (Basic Life Support) was averaged across state localities.

Other treatment costs

It was assumed that patients who require assistance from another person but do not require medical attention (SHE1) would receive a glucagon intramuscular or subcutaneous injection at a cost of \$55.02 per injection per event. This cost refers to the average price of a Glucagon Emergency Kit replacement vial. Prices are calculated from a payer perspective as the net effective sales price after 20% reduction from ex-manufacturer price due to discounts and rebates (which range between 15–25%)²⁶. The ex-manufacturer price was calculated as 80% of the net wholesale price of \$85.94 based on IMS unpublished data. The hypoglycemia model was designed to allow the option of including direct costs from a non-payer perspective. Therefore, costs for self-treatment following NSHE based on Brod *et al.*⁶ can also be included.

All costs applied in the economic model that predict the direct medical costs from a payer perspective associated with hypoglycemia are presented in Table 1, which also includes costs associated with out-of-pocket (OOP) expenses following self-treatment required to manage NSHEs. Brod *et al.*⁶ reported monthly OOP costs following self-treatment of \$11.50 for extra/special groceries, \$4.81 for food and drink from vending machines or café/restaurants, and \$4.73 for glucose products, such as tablets, gel tubes, or gum. Considering an event rate of 2.28 NSHEs per patient per month, this translates into OOP cost of \$9.23 per episode. Although excluded in the base case analysis, OOP expenses were additionally considered to

contribute to NSHE costs in sensitivity analysis. An outline of the approach taken for the direct medical cost calculation is shown in Table 2.

Frequency of resource requirements to treat hypoglycemia

Data on the frequency of resource items required to treat hypoglycemia events were obtained from three studies^{6,12,13}. For NSHEs, the frequency of resource requirements was based on data published by Brod *et al.*⁶, who analyzed the impact of NSHE on work productivity and diabetes management in 1404 individuals across four countries (US, UK, Germany, France). The author reported that 13.7% of US individuals contacted a HCP (either PCP, hospital, diabetes clinic, or other healthcare worker) as a result of the NSHEs, such as discussing dose adjustments⁶. In our economic model, we assumed the HCP contact corresponds to a physician visit for a medium, established patient.

Furthermore, Brod *et al.*⁶ reported that on average in the US population, each NSHE lead to an additional 3.9 SMBG tests. Although the study included patients with both T1DM and T2DM, the reported results were not stratified by diabetes type and, hence, extracted frequency data was applied equally to event cost calculations for individuals with T1DM and T2DM.

For SHE1, in the base case analysis, it was assumed that all individuals receive an intramuscular or subcutaneous glucagon injection. This was based on a previously reported costing approach for events requiring assistance by another person, but no medical attention¹⁰. Acknowledging that in clinical practice glucagon should be given only if patients are unconscious or unresponsive and unable to ingest oral glucose, lower frequency assumptions (75%, 50%, and 25%) were explored in sensitivity analyses as more conservative assumptions. Furthermore, HCP and SMBG costs for SHE1 were attributed to the same outcomes as those considered for NSHEs. Since no SHE1-specific data were identified in the literature review for the proportion of individuals that contacted a HCP and the number of additional SBMG tests subsequent to an event, we conservatively assumed the same data as applied for NSHEs, i.e., 13.7% contacted a HCP subsequent to a SHE1 event and 3.9 additional SBMG tests were assumed to follow each SHE1⁶.

The frequency of required services following SHE2 is based on the resource utilization described by Bullano *et al.*¹², who retrospectively analyzed 2315 patients with T2DM from the administrative claims database of a southeastern US managed care plan. From a total of 410 hypoglycemic events (100%) that required assignment of a hypoglycemia ICD-9-CM code (i.e., required HCP assistance), 20% were treated in an outpatient practice,

Table 2. Outline of the direct medical cost calculations for NSEH, SHE1, and SHE2 (T1DM and T2DM).

Procedures/medical interventions	Resource frequency				Resource cost		Resource cost weights		
	SHE2		SHE1		Cost	Reference	SHE2	SHE1	NSHE
	%	Reference	%	Reference					
Secondary care ER visit	17.00%	Bullano <i>et al.</i> ¹²	0.00%		\$139	Medicare 2012 Physician Fee Schedule + Addendum B of OPSS	\$24	\$0	\$0
Ambulance	21.32%	Leese <i>et al.</i> ¹³	0.00%		\$214	2012 Medicare ambulance fee schedule	\$48	\$0	\$0
Inpatient	0.00%		0.00%		\$4171	IPPS	\$0	\$0	\$0
Inpatient with ER	24.00%	Bullano <i>et al.</i> ¹²	0.00%		\$4171	IPPS	\$1001	\$0	\$0
Outpatient	20.00%	Bullano <i>et al.</i> ¹²	0.00%		\$325	Curkendall <i>et al.</i> ¹⁴	\$65	\$0	\$0
Primary care Visit to GP	26.00%	Bullano <i>et al.</i> ¹²	13.70%	Brod <i>et al.</i> ⁶	\$51	US Medicare 2012 Physician Fee Schedule	\$13	\$7	\$7
Home visit from GP	0.00%		0.00%		\$59		\$0	\$0	\$0
Nurse	13.00%	Bullano <i>et al.</i> ¹²	0.00%		\$43	85% of the physician visit based on US Medicare 2012 Physician Fee Schedule	\$6	\$0	\$0
Telephone contact with medical care	0.00%		0.00%		\$19		\$0	\$0	\$0
Other treatment I.M. glucose injection	0.00%		100.00%	Jönsson <i>et al.</i> ¹⁰	\$55	Average price of a Glucagon Emergency Kit replacement vial	\$0	\$55	\$0
Number of extra SMBG tests	3.90*	Brod <i>et al.</i> ⁶	3.90*	Brod <i>et al.</i> ⁶	\$1.05	Yeaw <i>et al.</i> ³⁴	\$4	\$4	\$4
post episode									
OOP per episode	0%		0%		\$9.23	Brod <i>et al.</i> ⁶	\$0	\$0	\$0
Total direct costs*							\$1161	\$66	\$11

ER, emergency room; GP, general practitioner; HCP, healthcare practitioner; SMBG, self-monitoring of blood glucose; OOP, out-of-pocket; OPSS, Medicare Outpatient Prospective Payment System; IPPS, Medicare Inpatient Prospective Payment System; I.M., Intramuscular.

*Number of extra SMBG tests per event.

*Total direct event costs = $\sum$ (resource frequency (i) $\times$ resource cost (i)).

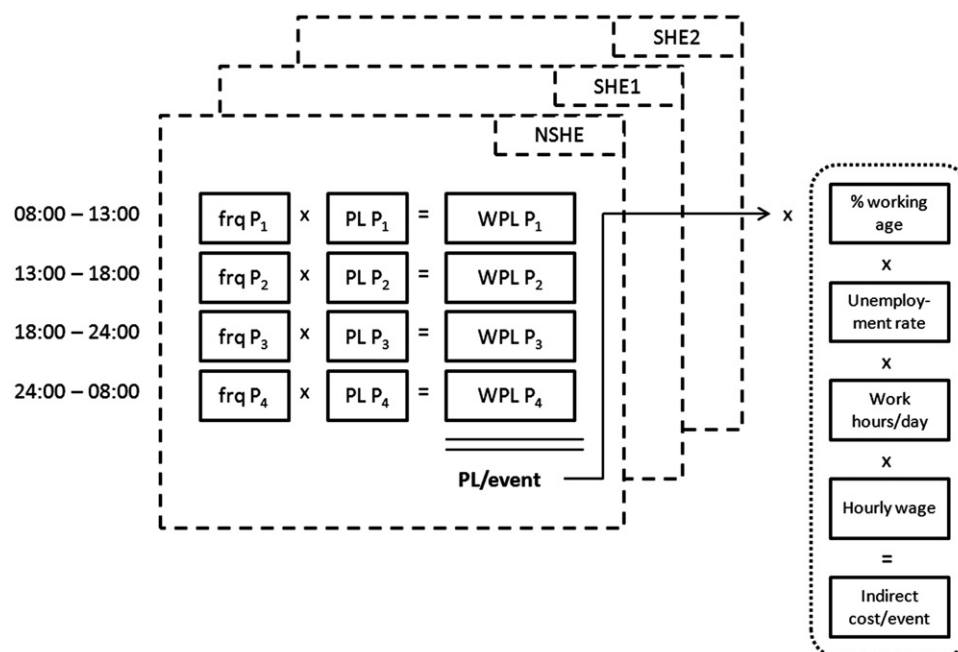


Figure 2. Indirect cost calculation model. P_1 to P_4 = four day and night time periods; $frq P_1$ to P_4 = frequency during which events occur across periods; PL = productivity loss; WPL = weighted productivity loss; $PL/event$ = total productivity loss per event.

26% were treated in office visits, 24% resulted in hospitalization including ER visit, 17% were treated in ER without subsequent hospitalization, and 13% resulted in nursing home encounters¹². These HCP services were regarded as mutually exclusive events. Office visits were considered to be equivalent to GP visits. The proportion of individuals requiring ambulance service following SHE2 was based on data from Leese *et al.*¹³, who reported data from 160 individuals with diabetes who experienced 244 episodes of severe hypoglycemia, requiring emergency assistance from a UK hospital unit. Of these, 89 (34%) involved contact with the ambulance service only, 19 (7%) involved accident and emergency/primary care services only, and 134 (55%) involved both. Accordingly, it was assumed that 55% of patients entering the ER or admitted to inpatient care with a prior ER visit require an ambulance¹³. Thus, if 24% of all SHE2 lead to hospital admission with prior ER visit and 17% are treated in ER with no subsequent hospitalization, it was concluded that in total 23% ($=55\% \times [24\% + 17\%]$) of individuals with SHE2 required ambulance services. As with the approach taken for SHE1, data from Brod *et al.*⁶ were adopted to assume an additional 3.9 SMBG tests followed each SHE2.

Indirect cost calculation

Indirect costs were defined as the expected productivity loss associated with hypoglycemia of different severity grades and estimated using a human capital approach. The human capital approach defines the value of a

human life as the discounted market value of the output produced by an individual over an expected lifetime. Estimates of productivity loss took into account the time at which the event occurred throughout the day. According to the approach taken by Gold *et al.*²², four time periods were defined: two work time periods (08:00–13:00, 13:00–18:00), after work (18:00–24:00), and nocturnal (00:00–08:00). The frequencies of hypoglycemic events occurring during each time period were obtained from the published literature^{6,22}, multiplied by the period-specific productivity loss, and totalled to yield the overall expected loss in productivity for one event. This was adjusted for the percentage of the population of working age and the unemployment rate to derive the net productivity loss per event (expressed in days). Finally, the productivity loss was converted to monetary units by multiplying the number of work hours per day by the average hourly wage, with the latter being derived from the annual income divided by the total working hours per year. Assumptions on hypoglycemia-related productivity loss and the frequency of episodes across daytime periods were extracted from two studies^{6,21}. The approach applied to the indirect cost calculation is schematically illustrated in Figure 2.

NSHE

Period-specific productivity loss assumptions for NSHEs are based on data from Brod *et al.*⁶, who reported that NSHEs experienced during periods 08:00–13:00, 13:00–18:00, 18:00–00:00, and 00:00–08:00 resulted on average

in 10.2, 10.2, 11.1, and 14.3 h of lost productivity, respectively. In that study, not all events resulted in productivity loss. The percentage of all NSHEs that resulted in lost productivity for the four consecutive periods was 10.79%, 10.79%, 8.47%, and 14.15%, respectively. Utilizing this data, the net amount of lost productivity for any NSHE was calculated as the product of work time lost per period multiplied with the percentage of individuals missing work time, which resulted in 1.10, 1.10, 0.94, and 2.02 h, respectively, of lost productivity for the four consecutive periods (e.g., $10.79\% \times 10.20 \text{ h} = 1.10 \text{ h}$). The hourly figures were divided by 8 h of average work time assumed per day for one individual to convert to daily units, leading to 0.14, 0.14, 0.12, and 0.25 days of lost productivity, respectively.

The frequency during which NSHE occur across the four time periods was obtained from the same study, which reported that, from a total of 790 NSHEs that had occurred in the US population, 278 occurred during work time (08:00–13:00 and 13:00–18:00), 307 outside work time (18:00–00:00), and 205 during night time (00:00–08:00)⁶. This leads to percentages of 17.6%, 17.6%, 38.9%, and 25.9%, respectively, for NSHEs occurring on average during the defined periods.

Period-specific productivity loss for the four consecutive periods was multiplied with the period-specific frequency weights and totalled to yield the overall expected productivity loss per episode. This number was finally adjusted for a 91.9% employment rate²⁷ and the presumption that 63% of the considered population are of working age (18–65 years)^{28,29}, and converted to monetary units considering an average annual income of \$27,915³⁰ at 40 work hours during the week and 48 working weeks in 1 year³¹.

SHE

According to data published by Gold *et al.*²², it was assumed that 16% of SHEs (both SHE1 or SHE2) occur between 08:00–13:00, 9% between 13:00–18:00, 35% between 18:00–00:00, and 40% during night time between 00:00–08:00 (i.e., nocturnal). Data for period-specific productivity loss following SHE1 and SHE2 were obtained from Davis *et al.*²¹, who reported lost productivity following severe events, mild-to-moderate events (which were combined in the study), and nocturnal episodes to be 3.6, 2.9, and 1.6 days, respectively, for individuals with T1DM, and 8.6, 2.7, and 2.5 days, respectively, for individuals with T2DM. The study did not further classify nocturnal episodes as mild, moderate, or severe. Productivity loss for mild-to-moderate events was applied to SHE1 in our calculation and respective data for severe events was applied to SHE2. Regarding the differentiation across day and nocturnal periods, Davis *et al.*²¹ differentiated productivity loss

only between daytime and nocturnal episodes. In our calculation, we applied the data for daytime episodes to the three periods between 8:00–0:00 and productivity loss following nocturnal episodes to the 0:00–8:00 period.

Likewise to the approach taken for NSHEs, the per-event productivity loss following SHEs was subsequently adjusted for employment rate and percentage of working-age population and then converted to monetary units.

Annualized direct and indirect costs of hypoglycemia

To put the direct and indirect costs of hypoglycemia into perspective, NSHE and SHE costs were applied on the hypoglycemia incidence reported by the UK Hypoglycemia Study Group⁵. That study reported incidence rates for SHEs of 10 and 70 events per 100 patient-years for individuals with T2DM treated with insulin for less than 2 years or more than 5 years, respectively. Respective rates for NSHE were reported at 410 and 1020 events per 100 patient-years. Following data from Leiter *et al.*³², we assumed that 11.8% of all SHEs are attributable to SHE2 and the remaining percentage to SHE1.

Health-related quality-of-life

A reasonable body of evidence exists informing the impact of hypoglycemia on health-related quality-of-life (HRQoL)^{3,21,33}. However, effects on HRQoL were not considered for the purposes of the hypoglycemia model. All costs are estimated in \$US and presented in 2012 values.

Results

Direct cost of hypoglycemic episodes

The direct cost calculation of hypoglycemic episodes is illustrated in Table 2. Direct medical costs associated with NSHEs were calculated as the sum of cost weights for the physician visits and additional SMBG tests and amounted to a total of \$11.09 per episode. This was based on cost weights of \$7.00 ($13.7\% \times \51.07) for the physician visit and \$4.10 for additional SMBG tests ($3.9 \text{ tests} \times \$1.05 \text{ per test strip}^{34}$). When OOP costs following NSHE self-treatment were included in the direct cost calculation, episode-related costs increased from \$11.09 to \$20.32.

Direct medical costs associated with SHE1 were calculated as the sum of cost weights for glucagon injection, HCP contact, and additional SMBG tests following SHE1. This resulted in a total direct medical cost of $\$55.02 + \$7.00 + \$4.10 = \66.11 for each SHE1. Sensitivity analyses assuming reduced glucose injection

frequency of 75%, 50%, and 25% resulted in total direct medical SHE1 costs of \$51.33, \$36.55, and \$21.78, respectively.

Resource costs weights for SHE2 were attributable to ER visits with no subsequent hospitalization, ambulance service, hospitalization with or without prior ER visit, outpatient visits, GP visits, nurse assistance, and extra SMBG tests and amounted to \$23.65, \$48.29, \$1001.04, \$64.96, \$13.28, \$5.64, and \$4.10, respectively. The sum of cost weights for these resources resulted in a total direct medical cost of \$1160.96 for each SHE2 (Table 2).

Indirect costs of hypoglycemic episodes

The overall expected productivity loss following NSHE was based on period-specific productivity loss and related frequency data and estimated at 0.16 days per NSHE (0.14 days $\times$ 17.6% + 0.14 days $\times$ 17.6% + 0.12 days $\times$ 38.9% + 0.25 days $\times$ 25.9% = 0.16 days). This translated into an indirect cost of \$10.75 for individuals with T1DM and T2DM (0.16 days $\times$ 91.9% $\times$ 63% $\times$ \$27,915/1920 work hours per year $\times$ 8 work hours per day) (Table 3).

Productivity loss following SHE1 episodes for individuals with T1DM was calculated as 60% $\times$ 2.9 days (weighted productivity loss for three diurnal periods) + 40% $\times$ 1.6 days (weighted productivity loss for nocturnal period) = 2.38 days and likewise for T2DM was calculated as 60% $\times$ 2.7 days + 40% $\times$ 2.5 days = 2.62 days. Following adjustment by employment rate, the percentage of working-age population and conversion to monetary units, indirect costs for SHE1 episodes were at \$160.27 for individuals with T1DM and \$176.43 for individuals with T2DM.

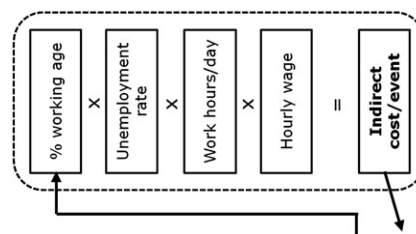
For SHE2 episodes, no difference was assumed between daytime and nocturnal productivity loss, and, hence, productivity loss reported by Davis *et al.*²¹ for severe episodes was directly applied (3.6 days for T1DM and 8.6 days for T2DM). This lead to episode-specific indirect costs of \$242.43 and \$579.14 for individuals with T1DM and T2DM, respectively. All data applied in the indirect cost calculation and a schematic illustration of the costing approach are presented in Table 3.

Annualized direct and indirect costs of hypoglycemia

Based on annual event rates from the UK Hypoglycemia Study Group⁵, direct annual per patient costs of hypoglycemia were estimated at \$45.48, \$5.83, and \$13.70 for NSHEs, SHE1s, and SHE2s, respectively, in individuals with T2DM for less than 2 years and \$113.13, \$40.82, and \$95.90, respectively, for individuals with T2DM for more than 5 years. Indirect costs following productivity loss were estimated to be \$44.08, \$15.56, and \$6.83 for

Table 3. Indirect cost calculation.

	NSHE (T1DM & T2DM) Brod <i>et al.</i> ⁶		PSF (SHE) Gold <i>et al.</i> ²²		SHE1 T1DM Davis <i>et al.</i> ²¹		SHE1 T2DM Davis <i>et al.</i> ²¹		SHE2 T1DM Davis <i>et al.</i> ²¹		SHE2 T2DM Davis <i>et al.</i> ²¹	
	PSPL	WPL	PSPL	WPL	PSPL	WPL	PSPL	WPL	PSPL	WPL	PSPL	WPL
08:00–13:00	0.14	0.024	16.0%	0.464	2.9	0.464	2.70	0.432	3.6	0.576	8.6	1.376
13:00–18:00	0.14	0.024	9.0%	0.261	2.9	0.261	2.70	0.243	3.6	0.324	8.6	0.774
18:00–00:00	0.12	0.046	35.0%	1.015	2.9	1.015	2.70	0.945	3.6	1.260	8.6	3.010
00:00–08:00	0.25	0.066	40.0%	0.640	1.6	0.640	2.50	1.000	3.6	1.440	8.6	3.440
Productivity loss/event (days)*		0.16		2.38		2.62		3.60		8.60		
Indirect costs/event (USD)		\$11		\$160		\$176		\$242		\$579		



PSF, period-specific frequency; PSPL, period-specific productivity loss; WPL = PSF $\times$ PSPL; NSHE, non-severe hypoglycemic event; SHE1, severe hypoglycemic event requiring non-medical assistance; SHE2, severe hypoglycemic event requiring medical assistance; T1DM, type 1 diabetes mellitus; T2DM, type 2 diabetes mellitus.
*Productivity loss/event = $\sum$ WPL.

Table 4. Direct, indirect, and total annualized costs of hypoglycemic episodes in T2DM individuals (\$US).

	Direct costs	Indirect costs	Combined costs
Per event costs			
NSHE	\$11	\$11	\$22
SHE1	\$66	\$176	\$242
SHE2	\$1161	\$579	\$1740
Per patient annual costs			
T2DM with diabetes duration <2 years			
NSHE	\$45	\$44	\$89
SHE1	\$6	\$16	\$22
SHE2	\$14	\$7	\$21
T2DM with diabetes duration >5 years			
NSHE	\$113	\$110	\$223
SHE1	\$41	\$109	\$150
SHE2	\$96	\$48	\$144

NSHE, non-severe hypoglycemic event; SHE1, severe hypoglycemic event requiring non-medical assistance; SHE2, severe hypoglycemic event requiring medical assistance; SHE, severe hypoglycemic event requiring (any) assistance; T2DM, type 2 diabetes mellitus.

individuals with T2DM for less than 2 years and \$109.67, \$108.93, and \$47.84 for individuals with T2DM for more than 5 years, for NSHEs, SHE1s, and SHE2s, respectively. Per patient annual costs, as well as event-based costs following hypoglycemia with different severity grades in the US, are presented in Table 4.

Discussion

A significant body of evidence shows that severe hypoglycemic episodes are associated with considerable costs. Few identified studies examined the economic implications of NSHEs, and most were limited to examination of medical resources, but none provided episode-related costs that may be considered in health economic evaluations to compare alternative treatment strategies that may reduce or prevent the incidence of hypoglycemia. As such, this study aids to fill the gap of economic data informing the consequences of hypoglycemic episodes that do not require medical assistance. The approach to derive direct and indirect costs for hypoglycemia may be adapted to other country settings to inform hypoglycemia costs in other economies. Beyond the factors that contributed to the direct and indirect hypoglycemia cost estimation in our study, several additional considerations contribute to the burden of hypoglycemia that are typically not considered in health economic evaluations. A recent survey of T2DM patients treated with insulin and HCPs reported that 10% of patients changed their basal insulin regimen, 14% intentionally let their blood glucose increase, and 16% missed doses to avoid hypoglycemia⁷. In addition, 82% of HCPs reported considering NSHE risk when choosing type of insulin therapy, and 57% started patients on lower insulin doses if a risk of hypoglycemia was elevated⁷.

It is also of note that hypoglycemia has been recognized as a contributor to clinical inertia in intensifying therapy, impacting maintenance of euglycemia^{35,36}. Although our study focused on the direct and indirect cost implications of hypoglycemic episodes, effects related to decreased patient adherence or contribution to clinical inertia in treatment intensification were not reflected in our considerations, but may contribute largely to the clinical and economic implications of hypoglycemia. For this purpose, more research is required, especially in real-world clinical settings, to quantify such effects on a patient-level basis, which would enable an integration in health economic evaluations and, as such, allow a generalized and over-reaching comparison of alternative diabetes interventions with differential hypoglycemia safety profile.

Hypoglycemia episode costs in perspective

Several studies have analyzed costs of hypoglycemia on a per-event basis, although the majority of the literature focuses on SHE costs. Although SHE costs from this study appear to be similar to the predictions from other sources, inconsistencies in cost predictions may result from several factors, including a difference in the methodology for cost calculations and the definition of hypoglycemic episodes. The direct medical costs of SHEs requiring medical assistance (SHE2) of \$1161 as predicted by our approach compared to \$1404 (95% CI = \$570–\$2237) direct medical costs paid on average by a US-managed care plan for individuals with T2DM who were analyzed in 2004¹². The same study was referenced to inform the resource utilization frequency for SHE2 in our costing approach. The lower cost estimation as predicted by our approach is likely related to the fact that we referred to unit complication costs from the Medicare insurance program, which are generally lower than the average amounts paid by a managed care plan. Another US study that analyzed data from 1992–1998 examining patients with T1DM and T2DM treated with insulin, found that the mean cost of hypoglycemia requiring medical attention was \$1723 per episode (range = \$263–\$7154)³⁷. Previous research reported data from a retrospective analysis of US insurance claims among 2.4 million patients with T2DM¹⁴. In that study, hypoglycemia episodes requiring medical intervention were recorded, and overall episode costs were assessed to be \$1008. When costs were analyzed by treatment setting, the overall hypoglycemia costs for patients treated with insulin amounted to \$1451 and \$622 for patients taking oral anti-diabetic drugs. The reduced costs for hypoglycemia episodes in patients treated with oral anti-diabetic drugs were attributable to the fact that fewer episodes lead to hospitalizations or ER visits.

No sources were identified that reported direct medical costs following hypoglycemic episodes that do not require

medical intervention (SHE1 or NSHE). This justifies the objective of our study to address the gap in economic data informing the consequences of hypoglycemic episodes that do not require medical assistance.

Differentiation of episode costs for T1DM and T2DM

For the estimation of direct costs, this analysis has sought to characterize the resource utilization associated with hypoglycemic events stratified by their severity. Importantly, this analysis does not explicitly differentiate between hypoglycemia-associated direct costs specific to diabetes type. However, these differences are implicitly captured via the differential frequency and severity of events that are likely to occur in T1DM or T2DM or the specific glucose therapy prescribed. Little evidence exists to inform the expected differences in direct cost following hypoglycemia in patients with T1DM and T2DM on a per episode basis. The postal survey conducted by Davis *et al.*²¹ suggested that, although overall for both T1DM and T2DM, the use of primary care resources following hypoglycemia episodes increased with the frequency and severity of episodes, no considerable difference in the percentage resource utilization per episode was evident between individuals with T1DM and T2DM. Although there may be specific factors associated with levels of resource utilization (e.g., age, comorbidities, and duration), the published literature on hypoglycemia-related costs do not report stratified analysis. Consequently, our estimated per-hypoglycemic event costs may fail to capture the true variability of costs that are likely to occur in clinical practice.

Larger diabetes type-specific differences may exist for indirect episode costs because productivity loss is likely to differ among diabetes types due to the different age distribution and employment situation. In our calculation, this is considered in the estimation of indirect costs for SHE1 and SHE2, which are evaluated separately for both diabetes types. However, for NSHEs, the data regarded for the indirect cost calculation are based on a study that included individuals with T1DM and T2DM who did not have subsequent stratification by diabetes type⁶, which only allowed for a generic estimation.

Limitations

Although the literature review conducted to inform this study was not instigated as a formal systematic review, the process undertaken was, nevertheless, rigorous, scientific, and reproducible. The research question, criteria for study inclusion, and data extraction were clearly identified and reviewed by all co-authors. Furthermore, we have provided relevant information for the interested reader in the online

supplementary material. We believe that the findings from this review process represent a balanced and impartial summary of the evidence available, conditional upon the databases searched.

Within our approach taken for the indirect cost calculation following SHEs, it is recognized that the definitions of hypoglycemia applied by Davis *et al.*²¹ are not precisely the same as those applied in our study. However, the application of productivity loss data from this study in the framework of our indirect cost calculation represents a conservative approach because, for both mild-to-moderate and severe events, the definitions from Davis *et al.*²¹ describe less severe events as the grades defined for SHE1 and SHE2.

It has to be further acknowledged that Davis *et al.*²¹ presented event-related productivity loss in terms of days lost from paid employment, non-paid activities, and the number of days helped by friends with usual activities. In the context of our indirect cost evaluation, we interpreted these costs as loss from paid employment. In the absence of data to indicate the proportion of productivity loss due to paid activities, we conducted a sensitivity analysis to assume that 50% of productivity loss presented by Davis *et al.*²¹ is due to paid activities and 50% due to non-paid activities and aid from friends/family. Following this assumption, indirect costs following SHE1 and SHE2 decreased from \$160 and \$242 to \$80 and \$121, respectively, for patients with T1DM and from \$176 and \$579 to \$88 and \$290, respectively, for patients with T2DM. Hypoglycemia can also cause loss of work for the family members or colleagues at work, if they need to help a family member or colleague with severe hypoglycemia. This contribution is considered in the productivity loss estimation derived for SHE1 and SHE2²¹, but not for NSHE⁶.

The fees applied to direct medical costs were determined using the medicare fee schedule; however, this by law is the lowest rate of reimbursement. The cost to private insured patients/systems and uninsured patients typically are higher than for medicare patients.

In rare cases a patient may suffer permanent brain damage or die due to hypoglycemia, which was not considered in our analysis due to the infrequent nature of these events.

Conclusion

The results of this analysis demonstrate that SHEs and NSHEs are associated with considerable costs in terms of both medical resource consumption and productivity losses. Failing to account for this cost burden may underestimate the value of diabetes management strategies that have an effect on hypoglycemia risk.

Transparency

Declaration of funding

This work was a joint project funded by the Eli Lilly and Co., Indianapolis, Indiana, USA and IMS Health.

Declaration of financial/other relationships

VF, DG, and JP are employees of IMS Health, a research consultancy that receives funding from pharmaceutical companies. PM undertakes paid consultancy for IMS Health and also receives research funding from pharmaceutical companies. NV, BC, and KB are employees of Eli Lilly; BC and KB are shareholders of Eli Lilly. JME peer reviewers on this manuscript have no relevant financial or other relationships to disclose.

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